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Multimodal Urban Heat Island Mitigation via Knowledge Reasoning and Geospatial Optimisation

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ABSTRACT

Urban Heat Island (UHI) mitigation demands curated strategies that are both ecologically effective and spatially feasible. This paper presents a dual-mode framework to support data-informed urban cooling interventions, combining passive knowledge-driven recommendations with active geospatial Land Use/Land Cover (LULC) optimisation. The passive approach leverages a Hybrid Retrieval-Augmented Generation (RAG) pipeline that fuses dense and sparse retrieval over a curated corpus of peer-reviewed research and policy literature. This enables the generation of evidence-based, transparent, and domain-validated textual recommendations for generalised UHI mitigation strategies.

Complementing this, the active mitigation pathway integrates computer vision, graph-based learning, and evolutionary optimisation to modify LULC configurations of the region of interest for targeted cooling of upto 3°C. The workflow begins with a cooling efficiency model that quantifies the additional green cover for the user-defined temperature reduction. Subsequently, morphological image processing generates a diverse population of feasible LULC variants by simulating realistic vegetation growth of the estimated amount within feasible urban areas. Finally, a Genetic Algorithm (GA), guided by a Graph Neural Network (GNN) trained to predict UHI intensity from spatial graphs, evolves this population to find optimal layouts that maximise cooling, preserve spatial coherence, and minimise infrastructure disruption.

The entire framework is deployed as a web-based geospatial intelligence platform, featuring an interactive 3D visualisation engine for rendering GA-optimised landscapes, quantitative dashboards for assessing the mitigation effectiveness, and an integrated suite of insight tools offering city-to-city comparative analysis, country-wise hotspot identification, and automated report generation. Overall, this work presents an end-to-end, scalable solution that empowers urban planners, policymakers, and researchers with actionable intelligence to build more climate-resilient cities.

INDEX TERMS Urban Cooling, LULC optimization, Retrieval-Augmented Generation, Graph Neural Network, Genetic Algorithm, Geospatial Intelligence.

I. INTRODUCTION

Urban Heat Islands (UHIs) are localized urban areas that experience significantly higher temperatures than surrounding rural regions due to the absorption and retention of heat by impervious surfaces, diminished vegetation, anthropogenic heat emissions and urban morphology that traps heat [1], [2]. This thermal disparity can result in an increase of surface temperature to about 10–15°C, while air temperature differences typically range from 1–4°C, especially during nighttime hours [3], [4]. In Europe, a study in 93 cities attributes about 6,700 annual premature deaths to the intensification of UHI, with almost half the heatwave mortality during the 2003 European crisis linked to urban heat amplification [5], [6]. Recent assessments indicate that approximately 34 million people in 65 major U.S. cities are exposed to UHI intensities of 4°C (8°F) or more, with low-income and marginalized communities disproportionately affected [7]. More than 55% of the world's population now lives in urban areas, and this figure is projected to rise to 68% by 2050. As a result, the frequency and intensity of UHI impacts on public health, energy use, infrastructure, and environmental quality continue to grow [8], [9].

To combat these effects, passive mitigation strategies such as reflective materials, cool roofs, and urban greening have

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shown promising results. Green roofs have demonstrated surface temperature reductions up to 13°C, while high-albedo surfaces can reduce cooling loads by up to 15% [10], [11]. Moreover, urban trees offer shade and promote evapotranspiration-based cooling, which can reduce ambient temperatures by about 0.7–1.5 °C. When implemented on a large scale, these measures have the potential to prevent thousands of heat-related deaths each year [12]. However, disparities persist: urban areas in the Global South achieve only about half the UHI mitigation benefits seen in the Global North, with a cooling potential of 2.1 °C compared to 3.8 °C [13]. These findings underscore the need for equitable and spatially informed solutions. In response, this paper presents a geo-intelligent UHI mitigation framework that extends our previous work by integrating remote sensing, spatial analytics, and an evidence-based recommendation system to generate locally optimized actionable strategies for urban thermal resilience [14].

II. LITERATURE SURVEY

The phenomenon of UHI poses significant environmental and public health challenges. Recent research has progressed in multiple complementary directions, including LULC optimization for improved spatial allocation of urban functions, the deployment of green infrastructure as a nature-based cooling strategy, predictive modeling frameworks to quantify UHI dynamics, and the application of generative design methods to explore alternative urban morphologies with embedded mitigation potential. Verified contributions and limitations of recent studies are summarized in the following.

A. EXISTING WORK

Generative Adversarial Networks (GANs) have been applied for LULC data augmentation and predictive modeling. A recent study [15] integrated remote sensing with GAN-based augmentation, improving the classification accuracy of zones affected by the UHI phenomenon, compared to conventional approaches. These methods improve detection efficiency, but mitigation remains indirect, as thermal outcomes are not explicitly modeled.

Green infrastructure provides measurable cooling effects. Marando et al. [16] analyzed 601 functional urban areas in Europe and found that Urban Green Infrastructure (UGI) reduces summer surface temperatures by an average of 1.07°C, with maximum reductions of up to 2.9°C. They showed that at least 16% tree cover is needed for 1°C cooling. Their regression models ($R^2 \approx 0.44$) predicted vegetation cooling effects, although errors of 0.5 highlight uncertainty.

Conceptual studies have explored various mitigation strategies. The research by [17] emphasizes the role of tree canopy, cool roofs, and green walls in reducing local temperatures, while Wang et al. [18] introduce Walkable Urban Cool Spots (WUCS), which integrates pedestrian walkability with outdoor thermal comfort by optimizing shading, vegetation, and surface materials.

Heat Risk Indices (HRI) guide intervention prioritization. According to [19], an HRI that combined LST, vegetation and population density found that 33% of the city was influenced by UHI, with hotspots concentrated in dense areas deprived of vegetation. Although effective for risk targeting, the index remains heavily dependent on LST and ignores socioeconomic and built-form variability.

Urban planning frameworks that integrate predictive modeling and optimization show further potential. Recent work by [20] integrated Local Climate Zone (LCZ) classifications with GAN-based prediction and optimization methods such as GA and Particle Swarm Optimization (PSO). Their review highlights improved predictive accuracy and optimized LULC allocations, although high computational costs limit real-time application.

Generative methods for urban design have also emerged. Previous work [21] demonstrated text-to-GAN pipelines to generate design layouts from textual input, producing conceptual alternatives but without explicit integration of thermal metrics or LULC constraints. UrbanGenoGAN, proposed in [22], combines GAN, GA and Geographic Information System (GIS) for grid-based spatial planning. Although the approach is effective in enhancing land use diversity and reducing UHI, reliance on rigid grids reduces applicability to irregular cities.

Morphological optimization highlights the value of spatial form. Zhang et al. [23] applied Geographically Weighted Random Forest (GWRF) to evaluate the morphologies of green spaces, showing that irregular patches outperform regular ones in cooling, particularly in commercial areas. However, zoning regulations and spatial constraints were not considered, which limits practical adoption.

City-scale simulation studies reinforce surface-driven cooling relationships. The findings in [24] used Weather Research and Forecasting-Princeton Urban Canopy Model (WRF–PUCM) to test cool and green roofs during a 2008 Baltimore–Washington heat-wave, finding near-linear reductions in UHI. Green roofs under moist conditions matched the performance of cool roofs (*albedo* ≈ 0.44), although assumptions of uniform adoption and single event testing limit generalization.

Spatial optimization has also been coupled with socioeconomic trade-offs. Chen et al. [25] classified LCZs with World Urban Database and Access Portal Tools (WUDAPT) and optimized them in Changsha using a GA – Future Land Use Simulation (FLUS) model. The results demonstrated a 5.98% increase in open low-rise areas and a 5.2°C reduction in LST, although the results depend on the precision of classification and assumptions of the scenario.

Generative urban design frameworks scale down to the block level. UDGAN, presented in [26] integrates GANs with multi-objective optimization. Training in 2,125 layouts, it achieved $R^2 \approx 0.8$ and generated design alternatives in minutes, reducing the iteration time. However, reliance on a single-designer data set restricts stylistic diversity. Similarly, a conditional GAN framework applied in Boston [27] opti-

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mized building footprints for shading efficiency, but did not integrate vegetation or surface reflectivity.

Broader architectural applications also illustrate the potential of AI. As summarized in [28], machine learning has supported 3D building modeling, block layout generation through reinforcement learning, and energy-based parametric optimization, yet most implementations remain conceptual and site-limited.

Finally, regional reviews highlight context-specific variation. A national assessment [29] reports UHI intensities of 1–5°C, extremes of up to 8.3° C in Delhi and an increase of 16° C LST in Pune. Remote sensing indices such as LST, Normalized Difference Vegetation Index (NDVI), LULC, and Normalized Difference Built-up Index (NDBI) were central, and deviation modeling achieved 0.157°C better accuracy than earlier methods.

B. NOVELTY IN OUR APPROACH

The proposed system addresses UHI mitigation by overcoming the rigidity of rule-based methods and the narrow scope of single-modality models. It operationalizes a dual strategy methodology through complementary passive and active pathways. The passive pathway uses a hybrid RAG–LLM to generate locally sensitive, text-driven strategies derived from urban knowledge sources. The active pathway employs an optimized land use simulation with evolutionary algorithms to generate LULC variants that meet defined cooling targets. These variants are assessed with a hybrid GNN that captures spatial dependencies and predicts UHI indices at high resolution. A multimodal fusion mechanism integrates text-driven guidance with spatial simulations, ensuring coherence and interpretability. This design establishes a differentiated end-to-end framework for UHI mitigation that advances beyond earlier approaches.

III. METHODOLOGY

The proposed methodology addresses UHI mitigation system adopts a dual-strategy methodology that integrates both passive and active approaches to address thermal stress in urban heat islands, as shown in Fig. 1. Upon detection of a UHI, the system activates two distinct but complementary mitigation pathways: a passive approach using Retrieval-Augmented Generation (RAG) to provide scientifically-grounded generalised recommendations, and an active approach using Land Use/Land Cover (LULC) optimisation to model targeted, site-specific interventions. Together, these two strategies offer a comprehensive framework for UHI mitigation. By integrating textual knowledge with geospatial intelligence, the system supports both policy-level planning and fine-grained urban design interventions to mitigate the UHI effect.

A. PASSIVE MITIGATION STRATEGY THROUGH TEXTUAL RECOMMENDATIONS

The passive mitigation strategy focuses on providing evidence-based, broadly applicable recommendations to lower surface temperatures and enhance thermal comfort. It

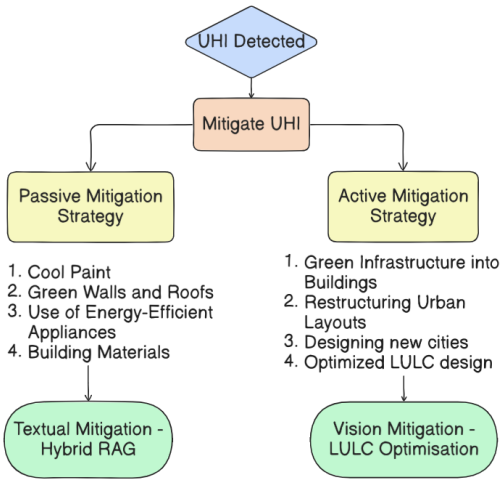

FIGURE 1. Mitigation Strategies

is powered by a custom Hybrid Retrieval-Augmented Generation (RAG) pipeline built on a curated corpus of scientific papers, technical reports, and policy documents related to UHI mitigation. By combining sparse (keyword-based) and dense (semantic-based) retrieval methods, the system ensures both precise and contextually relevant information retrieval.

This hybrid approach enables the model to generate scientifically grounded, trustworthy suggestions tailored to specific urban contexts. Common recommendations include using cool paints to reflect sunlight, installing green roofs and walls, creating shaded public spaces, adopting energy-efficient appliances, and promoting sustainable building materials. These interventions require minimal infrastructure changes, making them practical and adaptable across diverse urban environments.

To implement passive mitigation through smart textual recommendations, a Hybrid Retrieval-Augmented Generation (RAG) pipeline is created that employs a mixture of dense and sparse retrieval, and a retrieval-based answer generated by a large language model. The system contains the following components:

1. Deployment and Orchestration Stack

- Python to create the primary programming language for its compatibility with ML and NLP libraries.
- FastAPI to create a lightweight RESTful interface for preforming queries and obtaining results.
- Docker containers to allow reproducibility for ensuring compatibility across deployment modalities (including using it on Hugging face Spaces).
- The complete pipeline is orchestrated by LangChain, which managed component interaction (i.e. retrieval + generation), prompt formatting, memory management, and fallback logic.

2. Retrieval Layer

- Dense Retrieval - For the retrieval, user queries were embedded using the all-MiniLM-L6-v2 in the Hugging-

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Face Transformers, resulting in low dimensional dense vectors that maintain the same semantic relationships as the original query.

- Sparse Retrieval - In-tandem, BM25 Encoder from pinecone-text was used to create the sparse representations, which allows term-frequency based keyword matching.
- Hybrid Search - Each vector type is indexed in Pinecone. Pinecone has a hybrid retrieval capabilities built-in with the PineconeHybridSearchRetriever. With the hybrid search capabilities, the system permits the combination of semantic representations with algorithmically provided keyword matching.

3. Document Corpus

- Document corpus is the indexed corpus of peer-reviewed research articles, government guidelines, and academic reports related to UHI mitigation that has been preprocessed as chunked document items with metadata.

4. Generative Component

- Top-k retrieved documents is passed as additional context to the LLaMA large language model, which was hosted and accelerated via Groq API, for final answer generation using the context window provided.

5. Response Formatting

- Generated output is encapsulated into a JSON response (structured) before being returned to the user through a FastAPI endpoint. This will include: synthesized answer, references to source documents, and optional confidence levels.

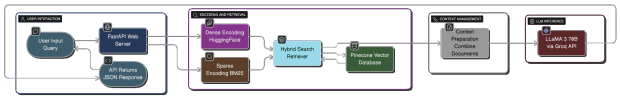

FIGURE 2. Architectural representation of the RAG deployment

B. ACTIVE MITIGATION VIA LULC OPTIMISATION

The active mitigation strategy employs a data-driven optimisation framework, illustrated in Fig. 3, to generate optimal LULC modifications for targeted UHI reduction. The process is initiated with two primary inputs: the current LULC map of the region of interest and a user-defined cooling objective, such as a desired reduction in the temperature by x°C. Using this target, a quantitative pre-processing module estimates the required percentage of new green cover ($k\%$) that must be added to this location to achieve the desired level of thermal mitigation. The system then generates a diverse population of feasible LULC variants. Each variant represents a unique spatial configuration for allocating the target $k\%$ green cover onto the existing LULC map covering a wide variety of placement strategies, like clustered, dispersed, and corridor-like patterns. To ensure the generation of realistic and implementable recommendations, the optimisation is governed by a set of predefined constraints. The model performs a pixel-wise semantic analysis to identify permissible areas for modification, treating critical infrastructure, water bodies, and other user-defined zones as immutable. Subsequently, a GA optimises this population to identify the optimal LULC configuration. It iteratively evolves the population of feasible LULC variants, evaluating each candidate against a multi-objective fitness function designed to maximise the predicted cooling effectiveness based on the spatial arrangement of the new green infrastructure and minimise disruption to existing built-up areas and critical infrastructure, yielding a final prescriptive map that guides the most effective, data-driven placement of green infrastructure to meet the initial cooling goal while preserving urban functionality and spatial integrity.

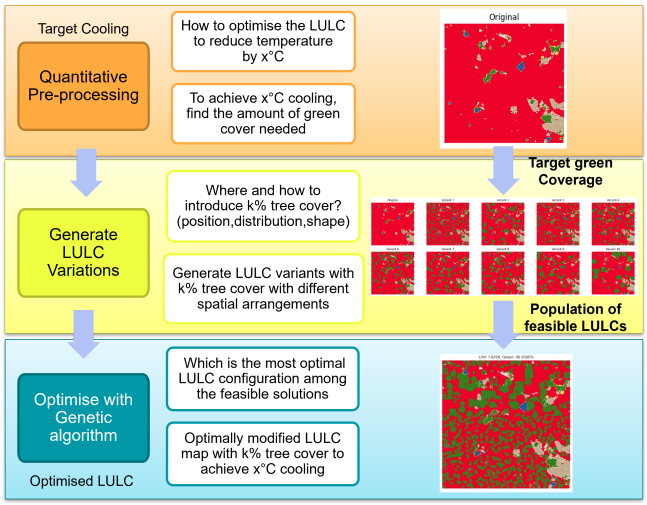

FIGURE 3. Workflow of the Active Mitigation Strategy using LULC Optimisation.

1) Quantitative Pre-processing

The first stage of the active mitigation workflow is to quantify the required increase in green cover, P_{req}, needed to achieve a user-defined target reduction in the temperature ΔT_{target}. Multiple empirical studies have established a strong, inverse relationship between green cover and LST, with higher percentages of vegetation correlating with reduced temperatures [30], [31]. This phenomenon is driven by the cooling effects of shading, evapotranspiration, and higher albedo caused due to green cover. This empirical finding is best represented by the linear model:

$$P_{req} = C \cdot \Delta T_{target} \tag{1}$$

where:

P_{req} is the required percentage increase in tree cover.
ΔT_{target} is the target LST reduction like 1, 2, or 3 °C.
C is the empirical cooling coefficient, representing the percentage of tree cover needed to achieve a 1 °C tem-

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perature reduction within the specific Functional Urban Area (FUA).

The coefficient C is a crucial parameter that encapsulates local climatic conditions, typical vegetation cooling efficiency, and background LST. Based on the established findings and regional empirical studies, it is set to $C \approx 16\%/°C$ [16]. Consequently, target reductions of 1°C, 2°C, and 3°C correspond to required tree cover increases of approximately 16%, 32%, and 48%, respectively. The calculated P_{req} provides the quantitative, area-based objective for the subsequent spatial optimisation stage.

2) Generating feasible LULC variants using morphological image processing

After determining the target percentage of green cover required for achieving a desired cooling effect, the system proceeds to generate a diverse set of spatially feasible LULC variants that incorporate the estimated amount of vegetation. This process leverages a series of image processing techniques to simulate realistic vegetation expansion over an urban landscape, while adhering to spatial and ecological constraints.

A binary mask is first constructed to identify all candidate pixels that are eligible for green cover addition. Fixed classes such as water bodies, snow/ice, and immutable infrastructure are excluded from modification, creating a constrained canvas where new vegetation can be introduced.

The generation of each LULC variant is an iterative, stochastic process. The system progressively adds vegetation patches to the modifiable areas until the target coverage of $k\%$ is achieved. The creation of each patch is a multi-step procedure designed to mimic natural growth:

1) Seed Point Selection: A patch originates from a randomly selected seed pixel within the eligible area.
2) Patch Growth via Morphological Dilation: The growth is simulated using the morphological dilation operator. A structuring element (SE) like a disk or square or any random shape of varying size is applied for a random number of iterations to expand the seed point into an amorphous, connected region. This technique effectively simulates the organic spread of vegetation from a central point, similar to how trees or shrubs grow outward from a cluster.
3) Post-processing and Refinement: To enhance realism, geometric transformations such as rotation, morphological closing (to fill small holes), and edge smoothing are applied. Finally, any generated patches falling below a minimum area threshold are discarded to prevent unrealistic micro-vegetation and ensure each patch is thermally significant.

Stochasticity is the core component that ensures diversity across the population of LULC variants. By randomising the seed locations, the shape and size of the SEs, and the number of dilation iterations, the system generates a high degree of variation in patch size, shape, and placement, as illustrated in Fig. 4. The final output is a diverse set of candidate solutions, each of which incorporates the required amount of green cover but with different spatial arrangements. These variants are topologically coherent (no abrupt transitions), ecologically valid (green only on permitted land types), visually realistic (patches mimic organic vegetation spread), and quantitatively constrained (match the target $k\%$ percentage of tree cover).

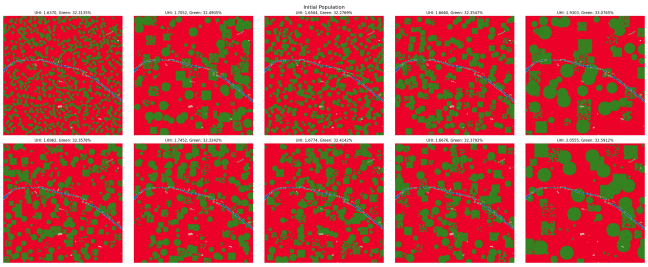

FIGURE 4. Generation of candidate LULC variants where each variant meets the target green cover percentage with a unique spatial configuration.

3) LULC Optimization via Genetic Algorithm

The feasible LULC configurations generated in the preceding stage constitute the initial population for the proposed Genetic Algorithm. To derive spatially optimal land-use distributions aimed at mitigating the UHI effect, a GA-based optimization framework is employed. The algorithm iteratively evolves the population of LULC variants through biologically inspired evolutionary operators—namely, selection, crossover, mutation, and elitism. The evolutionary process is guided by a composite fitness function formulated to enhance green cover, minimize UHI intensity, and preserve spatial coherence within the optimized land-use patterns.

a: Fitness Function Formulation for the Genetic Algorithm

Each individual in the initial population represents a unique LULC configuration . The fitness of each candidate is evaluated using a multi-objective fitness function that incorporates ecological, thermal, and spatial realism criteria. The fitness function is defined as:

$$\text{Fitness}_{GA} = K \cdot \Delta G + L \cdot \Delta B + N \cdot UHII + R \quad (2)$$

where:

- ΔG Net change in vegetated area relative to the parent LULC configuration.
- ΔB Net change in built-up area relative to the parent LULC configuration.
- $UHII$ Predicted Urban Heat Island intensity, inferred from LULC layout using the proposed Graph Neural Network architecture.
- R Realism coefficient, computed as the maximum radius of green-cover regions, penalizing spatially implausible vegetation insertions.

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K, L, N Empirically tuned weights that control trade-offs among greening, urban functionality, and heat mitigation.

b: Predicting UHII from LULC via Graph Neural Networks

To estimate the Urban Heat Island Index from spatial land-use data, a graph-based learning framework adopting Graph Neural Networks was developed. GNNs are particularly suitable for this task as they explicitly encode spatial relationships [32], enabling the model to capture interactions between heterogeneous land-cover regions and their neighbors. This spatially-aware representation allows for a more accurate modeling of how local land-cover compositions collectively influence urban heat distribution.

Graph Construction from LULC Maps:

The LULC map is transformed into a graph structure, where each spatial patch is represented as a node and spatial adjacency defines the edges. The process involves three key steps.

1) *Node Construction:* The LULC map is divided into equal-sized patches, each serving as a node. Every node is characterized by a feature vector encoding the proportion of land-cover types within that patch, thereby summarizing local landscape composition.
2) *Edge Construction:* Nodes are connected based on spatial proximity. Each node is linked to its immediate neighbors, forming a grid-like topology that preserves spatial continuity and allows the model to learn inter-patch dependencies.
3) *Graph Assembly:* For each region, the node feature matrix captures land-cover distributions across patches, the edge index encodes adjacency relations, and the corresponding UHI Index serves as the graph-level label.

A training dataset is created by pairing these LULC-based graphs with their corresponding UHII values (ranging from –5 to 10) from our previous work [14]. This complete dataset is then used to train the Graph Neural Network model.

Model Architecture:

The proposed GNN-based regression model, integrates a hybrid graph convolutional backbone with a regression head. The backbone consists of three sequential convolutional layers: SAGEConv, GATConv, and GCNConv. Each layer plays a distinct role: SAGEConv performs robust neighborhood feature aggregation, GATConv introduces anisotropic attention to focus on the most influential neighbors, and GCNConv acts as an isotropic smoother that regularizes the learned embeddings. ReLU activations follow each layer. Node embeddings are aggregated using a global mean pooling layer to produce a unified graph-level representation, which is then passed through a three-layer Multi-Layer Perceptron (MLP) to predict the final UHI value. The complete workflow of the GNN pipeline is depicted in Fig. 5.

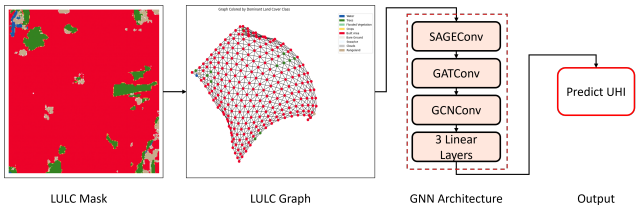

FIGURE 5. Complete GNN Workflow for LULC-based UHI Prediction

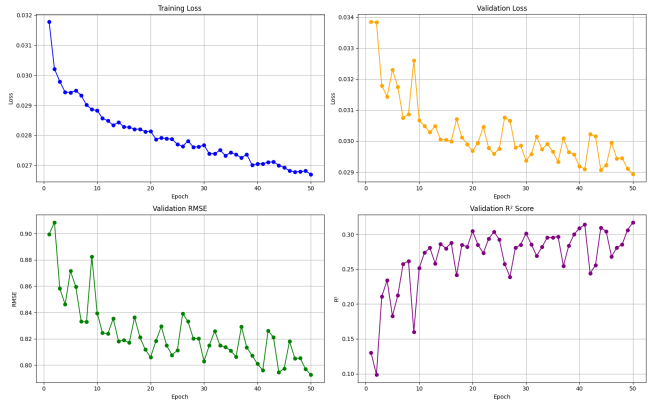

FIGURE 6. Training Curves of the proposed Graph Neural Network Architecture.

Optimization and Training:

The model is optimized using the Huber loss function with a threshold parameter $\delta = 0.05$. This loss was preferred over the conventional Mean Squared Error (MSE) due to its robustness to outliers in UHI measurements. Acting as a combination of Mean Absolute Error (MAE) and MSE, it behaves quadratically for small errors and linearly for large ones, preventing extreme UHI values from disproportionately affecting gradient updates. As shown in Fig. 6, the training curves depict the model's learning progression.

Results:

On the test dataset, the model achieved an R^2 score of 0.3870, a Root Mean Squared Error (RMSE) of 0.7926, and a MAE of 0.6040. Fig. 7 shows the prediction of the model for a couple of LULCs. This trained model is utilized to predict the UHII values for generated LULC masks, which is a critical step for computing the fitness function within the genetic algorithm.

c: Genetic Algorithm for Population Evolution

The evolution of the initial LULC population is driven by a Genetic Algorithm designed to optimize solution fitness over a predefined number of generations. Each generation cycle implements a sequence of selection, crossover, mutation, and replacement to progressively improve solution quality. Figure 8 illustrates the workflow of the Genetic Algorithm.

First, an elitism strategy is employed to ensure the preservation of high-performing solutions. The top 5% of individuals from the current population, sorted by fitness, are directly retained and passed into the new population. To maintain

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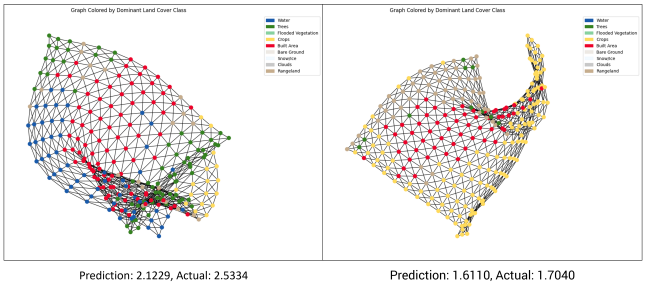

FIGURE 7. Inferencing the trained GNN to predict UHI Index from a given LULC Mask

genetic diversity and prevent premature convergence, an additional 5% of the population is selected randomly from the non-elite individuals and is also carried over.

The remainder of the new population is generated through reproductive operations. Parent individuals are chosen using a rank-based selection mechanism. In this method, individuals are ranked according to their fitness, and selection probabilities are assigned based on this rank rather than the absolute fitness scores. This stochastic approach balances selection pressure, mitigating the risk of premature convergence to suboptimal solutions by allowing less-fit individuals a probabilistic chance to reproduce.

Once two parents are selected, a single-point crossover operation is performed to generate a single child LULC mask. This offspring is then subjected to a critical validation and variation process based on its environmental impact, specifically the green increment, the change in green cover percentage compared to its parent.

This increment is evaluated against a dynamic threshold, which is linearly annealed from an initial value toward zero as the generations progress. This strategy imposes a stricter requirement for improvement in later generations. If the child's green increment is positive but falls below this dynamic threshold, a mutation operation is applied to the child mask to introduce new genetic material.

A final viability check is performed: if the green increment is negative, indicating a reduction in green cover, the child is deemed non-viable and is discarded. If the child is viable (i.e., green increment is non-negative), its fitness is evaluated, and it is added to the new population. This cycle of selection, crossover, and constrained mutation continues until the new population is fully constituted. This new generation then serves as the input population for the subsequent iteration of the algorithm.

d: Algorithm Outputs and Framework Validation

Upon completion of the predefined number of generations, the genetic algorithm yields a final, evolved population of candidate LULC scenarios. From this optimized cohort, the LULC design corresponding to the maximum fitness score is identified as the principal recommended solution.

Ultimately, this integrated framework, which couples a Cooling Index strategy for initial population generation with a multi-objective Genetic Algorithm, provides a robust methodology. It offers a powerful tool for generating and evaluating actionable urban redesign strategies specifically aimed at mitigating the adverse effects of the Urban Heat Island phenomenon.

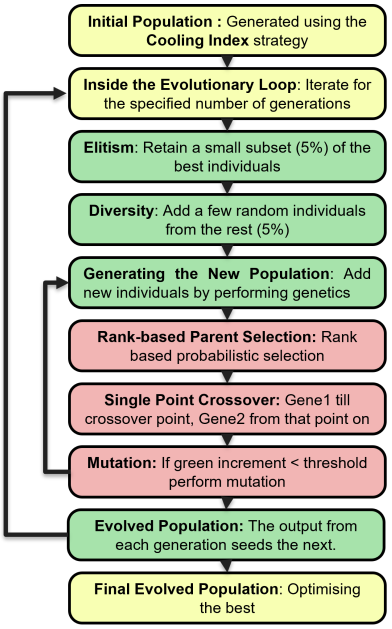

FIGURE 8. Evolutionary Workflow for LULC Optimization using a Genetic Algorithm

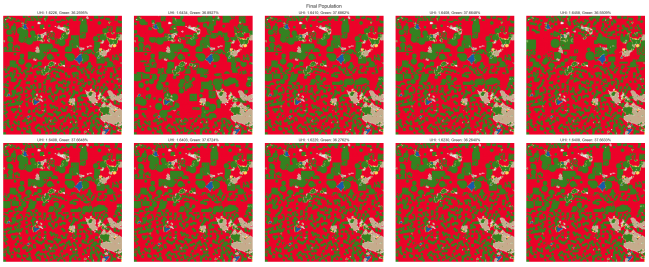

FIGURE 9. Final Optimized Population

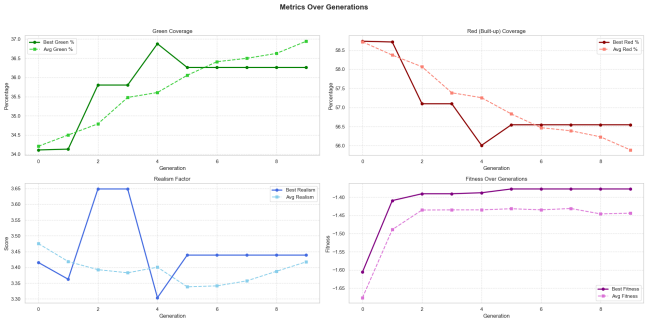

FIGURE 10. Variation of parameters like built area, green cover, Fitness function, realistic factor throughout evolution

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4) Interactive 3D Visualization of Optimised LULC

To facilitate an intuitive and immersive comparison between the existing urban landscape and the GA-optimised LULC, an interactive 3D visualisation system is developed. Standard 2D maps are often insufficient for communicating the vertical and spatial impact of urban design interventions, especially to non-technical stakeholders. This 3D viewer addresses this limitation by extruding 2D LULC rasters into interactive 3D column maps that can be viewed at all angles, as shown in Fig. 11.

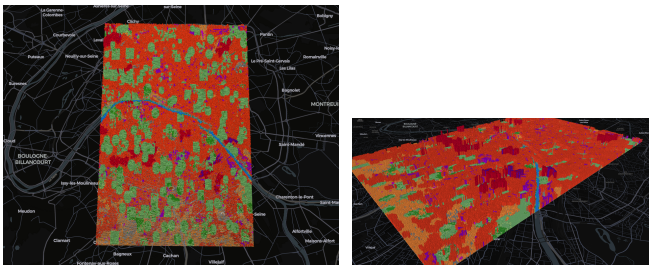

FIGURE 11. 3D rendering of the LULC in Paris

The foundation for the 3D scene is a 1024×1024 pixel high-resolution, multi-class map of the current urban environment, which is extracted from the Global Human Settlement Layer (GHSL) Morphological Settlement Zone (MSZ) dataset (GHS-BUILT-C R2023A), which provides 15 detailed classes of built-up and open spaces [33]. A Coordinate Reference System (CRS) transformation is performed on this LULC, reprojecting the patch from the dataset's native Mollweide projection (`ESRI:54009`) to a standard WGS84 projection (`EPSG:4326`) used throughout the analysis. The GA-optimised LULC map is compared with this ground truth LULC map to isolate and visualise the exact pixels altered by the GA. The final step is a 3D rendering that maps each of the final LULC classes to two key properties, colour and height, as listed in the Table 1. It converts each pixel into a geospatial point and renders it using `Pydeck`'s ColumnLayer. This layer extrudes each point vertically based on its assigned height and applies the corresponding colour. The final output consists of two interactive 3D models, the existing LULC and proposed optimised LULC, allowing urban planners and policy makers to intuitively explore and compare the spatial and vertical impact of the proposed UHI mitigation strategies.

IV. RESULTS

A. END-TO-END UHI MITIGATION ANALYSIS FOR A SINGLE ROI

This section presents the comprehensive mitigation analysis for the selected Region of Interest (ROI), Bengaluru, India. Based on our established quantification methodology in our previous work, the city's baseline UHII was determined to be 3.11°C [14].

Following this baseline assessment, the dual mitigation framework was applied. The passive mitigation approach, utilizing the hybrid RAG-LLM, produced a set of textual

TABLE 1. LULC Class Mapping for 3D Visualization

ID	Description	Rendered Ht. (m)	Color
	Open Spaces		
01	Low Veg. (NDVI $\leq$ 0.3)	3	Light Green
02	Med Veg. (0.3 $<$ NDVI $\leq$ 0.5)	5	Lime Green
03	High Veg. (NDVI $>$ 0.5)	10	Dark Green
04	Water	1	Blue
05	Road	1	Gray
	Built Spaces - Residential		
11	Building Height (Ht.) $\leq$ 3m	10	Pale Yellow
12	3m $<$ Building Ht. $\leq$ 6m	30	Gold
13	6m $<$ Building Ht. $\leq$ 15m	60	Orange
14	15m $<$ Building Ht. $\leq$ 30m	150	Red-Orange
15	Building Ht. $>$ 30m	350	Maroon
	Built Spaces - Non-Residential		
21	Building Ht. $\leq$ 3m	10	Light Pink
22	3m $<$ Building Ht. $\leq$ 6m	30	Pink
23	6m $<$ Building Ht. $\leq$ 15m	60	Magenta
24	15m $<$ Building Ht. $\leq$ 30m	150	Purple
25	Building Ht. $>$ 30m	350	Dark Purple

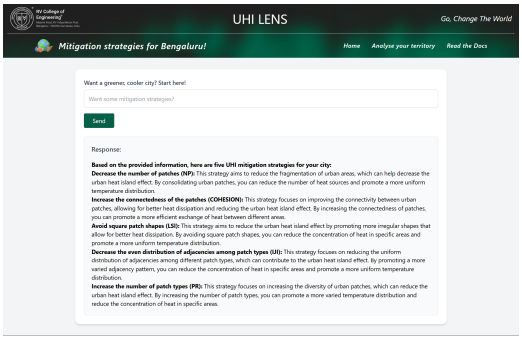

FIGURE 12. Passive Mitigation Strategy Recommendations for Bengaluru

recommendations encompassing specific policy and material-based strategies. These generated recommendations for Bengaluru are detailed in Fig. 12.

For the active mitigation strategy, an urban redesign was initiated to achieve a target cooling of 2°C. This process began by generating an initial population of LULC variants engineered to include a 32% increase in green cover. This population subsequently underwent an evolutionary optimization via the genetic algorithm. The resulting optimal solution—the LULC design exhibiting the highest fitness score—is presented in Fig. 13 (a) along with a comparitive 3D rendering of the current vs proposed LULC in Fig. 13 (b), which contrasts the city's current urban design with the proposed optimized landscape.

B. INSIGHTS TOOLSET

From region-specific evaluations to global comparisons, the website offers a set of interactive tools that improve usability and support UHI analysis. Researchers, planners, and policymakers can easily compare urban conditions, identify heat patterns, and develop effective strategies for mitigation.

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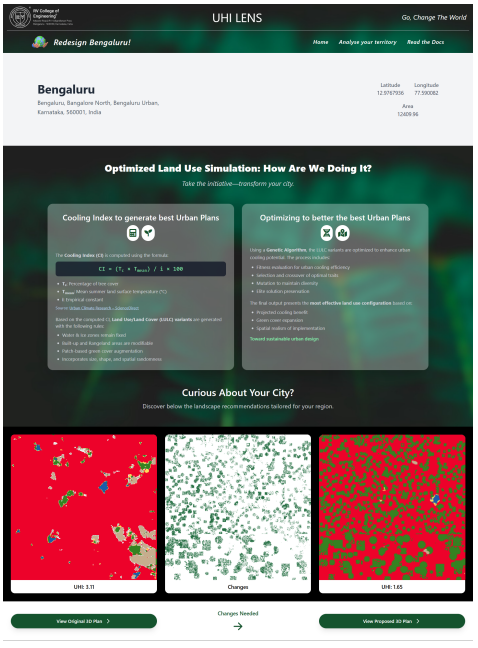
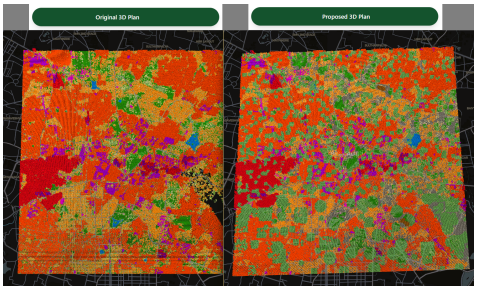


[a]



[b]

FIGURE 13. a) Active Mitigation via LULC Optimisation for Bengaluru, b) Side by side 3D analysis for existing vs proposed LULC

1) Timelapse

The Timelapse feature provides a detailed analysis of an ROI over a long period. It helps users understand the specific factors that drive the thermal changes instead of just seeing their effects. By visualizing how land use and land cover patterns and key Sentinel indices, such as NDVI and NDBI, change over the years, users can see how factors like reduced green spaces or increased built-up areas affect Urban Heat Island Intensity (UHII). This analysis supports urban planning based on data by helping identify the main causes of rising temperatures and predict future UHI scenarios using past data. It also guides planners in directing mitigation strategies, such as adding green infrastructure or cool pavements, to the areas that need them most. The analysis of Detroit shown in the Fig.14 clearly illustrates how the ROI changes over time.

2) Comparison Analysis of Two Cities

The Comparison Analysis feature as shown in Fig. 15 allows users to select any two cities and compare their UHI characteristics side by side. Satellite imagery, LULC maps, and UHI trends over time are presented together. This enables users to observe variations in heat intensity across different

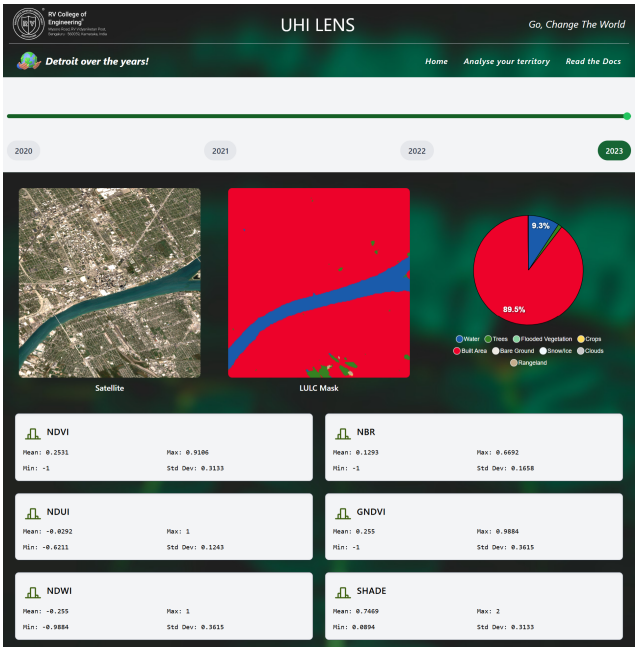


FIGURE 14. LULC and Sentinel features of Detroit across different years, navigated using a sliding toggle.

areas and over time. This comparative insight helps reveal patterns of environmental stress and thermal anomalies caused by urbanization. It highlights specific periods and locations with the highest heat exposure. Understanding these factors helps support targeted interventions, such as increasing urban vegetation or revising building layouts, to improve thermal comfort and overall urban resilience.

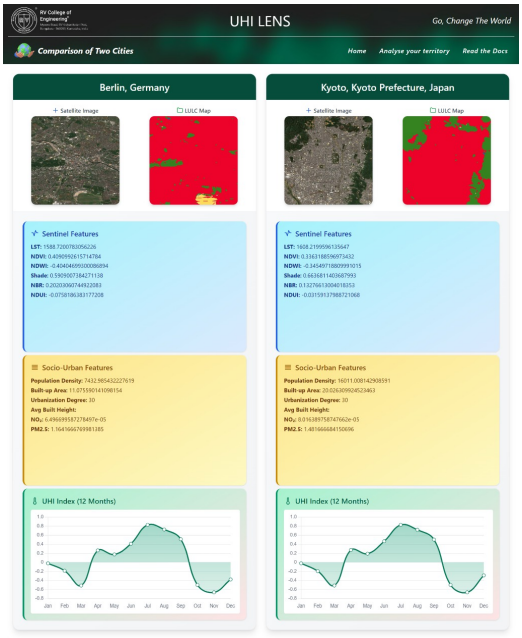


FIGURE 15. Comparison Analysis of two cities—Berlin (Germany) and Kyoto (Japan)

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3) Countrywise Look Up

The Country-wise Lookup feature, as displayed in Fig. 16 allows users to explore a nation's UHI situation by mapping its top 20 cities, each marked with coordinates and UHI intensity. The interactive map uses a color-coded scale to visually show heat stress levels. This helps users quickly identify national hotspots and emerging risks related to heat. By providing an overview of urban heat vulnerability, this tool aids in decision-making, efficient resource distribution, and prioritizing cities that need urgent mitigation efforts. It also highlights how land use practices and rapid urban growth contribute to heat accumulation on a larger scale.

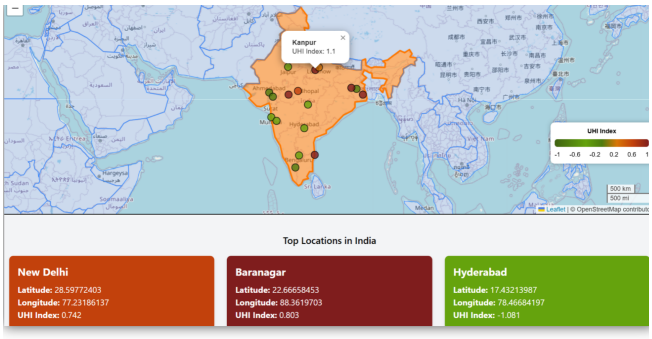

FIGURE 16. Country-specific hotspots analysis—India

4) WorldWide Analysis

The worldwide analysis feature provides a global perspective on UHI patterns by aggregating data across countries and continents. It visualizes heat intensity in major cities worldwide, highlights international hotspots, and enables comparisons of regional variations in urban thermal stress. The tool allows visualization of 15 Sentinel-based features (NDVI, Shade, NBR, NDUI, GNDVI, NDWI, NDBI, NMDI, NDBSI, NBaDI, WVP, AOT, LST, SAVI, UHSI) and 9 socio-urban features related to population density, degree of urbanization, infrastructure, and pollutants, as abbreviated and explained in our previous work [14]. The worldwide analysis feature is depicted in Fig. 17. For each feature, users can select the statistical layer (mean, minimum, maximum, or standard deviation) for visualization, enabling more detailed quantitative analysis across regions. By integrating satellite-based UHI indices with sentinel and socio-urban indicators, the tool reveals how urbanization patterns, climate zones, and land-use strategies influence heat accumulation on a global scale. This broader view supports cross-regional benchmarking and the development of coordinated strategies for heat mitigation and climate-resilient urban planning.

5) Report Generation

The report generation feature automates the creation of comprehensive UHI assessment reports using Python's ReportLab and Pillow libraries. Once the analysis for a selected location is complete, the system generates a structured, multipage PDF

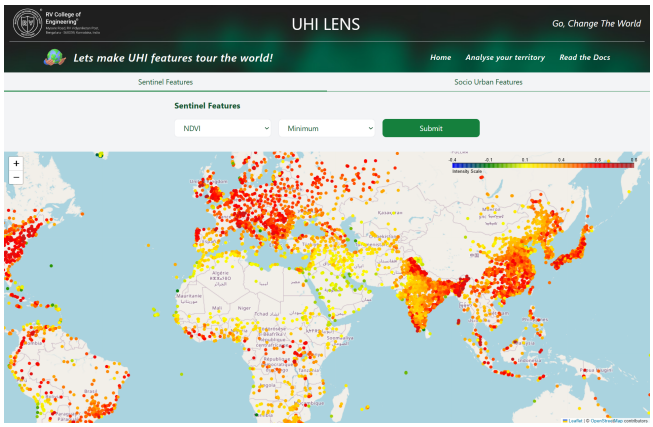

FIGURE 17. Geospatial Mapping of Urban Thermal Stress Across Major Cities Using Sentinel NDVI Metrics

that consolidates all analytical results and visual outputs into a clear, readable format, as illustrated in Fig. 18.

To provide a complete record of the heat analysis for the selected region, the report includes the following components:

- Overview of UHI and its significance
- Socio-urban profile of the city (9 key parameters)
- Sentinel-based indices (15 features with descriptive information and statistical summaries)
- UHI Index
- LULC map of the city
- Seasonal 12-month UHII variation trend
- Time-lapse visualization of the city's evolution over the past 4 years
- Mitigation strategies, including:
  - -- Active measures (original vs. proposed LULC and suggested changes)
  - -- Passive measures (LLM-generated RAG-based recommendations)
- Conclusion summarizing the key findings and recommendations

This structured report enables users to interpret localized heat stress effectively and supports data-driven decision-making for urban heat mitigation and resilient city planning.

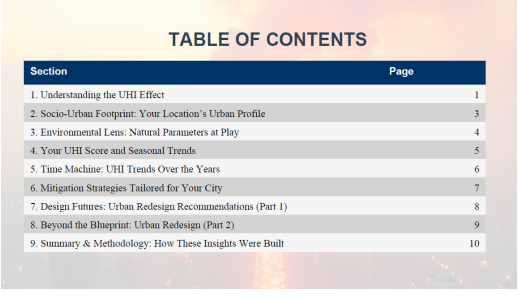

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FIGURE 18. Table of contents showing report sections

V. DISCUSSION

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1) Advantages:

The proposed framework presents a comprehensive and adaptable approach to Urban Heat Island mitigation by integrating passive textual recommendations through a Hybrid RAG system with active spatial interventions via optimization algorithms. This dual strategy delivers both immediate, evidence-based insights and long-term, spatially grounded solutions. The Hybrid RAG ensures scientific reliability by combining dense semantic and sparse keyword-based retrieval from a curated corpus of peer-reviewed research, while the genetic algorithm-based spatial optimizer generates realistic Land Use/Land Cover modifications that respect existing infrastructure and maximize cooling potential. The system's scalable design allows deployment across diverse urban settings, adaptable to local climatic and geographic conditions through empirically derived parameters. Additionally, its multi-objective optimization balances green cover enhancement, UHI reduction, urban functionality, and spatial realism. The integration of Graph Neural Networks further improves predictive accuracy by capturing complex spatial and microclimatic interactions often overlooked by traditional models. The proposed Genetic Algorithm-Graph Neural Network pipeline offers better interpretability compared to conventional Deep Learning models. It explicitly correlates specific land-cover modifications with the predicted UHI variations, making the rationale behind the spatial interventions clear.

2) Limitations:

The framework offers a solid foundation for Urban Heat Island (UHI) mitigation, but it has several limitations. The reliance on computationally intensive processes like genetic algorithm optimization and GNN training can limit real-time use in resource-constrained environments. The moderate predictive accuracy of the GNN model(R^2 = 0.387) indicates room for improvement by incorporating additional environmental variables or advanced architectures. Effectiveness also depends on the availability and quality of satellite imagery and LULC data, which can vary across regions. Current validation is limited to the Paris case study, so broader testing across diverse climates and urban forms is needed. Moreover, the system does not yet address practical implementation challenges such as zoning, funding, or stakeholder coordination, and its simplified thermal modeling and static data analysis overlook dynamic factors like microclimates, seasonal changes, and urban growth.

VI. CONCLUSION

This study introduces a unified framework for Urban Heat Island mitigation that combines remote sensing, machine learning, and natural language processing to enable both broad strategic planning and detailed, location-specific interventions. The passive component uses a Hybrid Retrieval-Augmented Generation pipeline to provide evidence-based recommendations, while the active component employs land use data, optimization algorithms, and Graph Neural Networks to design localized land cover modifications that help lower surface temperatures. An automated reporting module further organizes the results into a clear, actionable format for use in urban planning and policy. This integration of knowledge-driven and data-driven approaches demonstrates the potential for more climate-responsive urban design. Looking ahead, the framework can be enhanced by incorporating dynamic environmental variables, 3D urban morphology, and high-resolution temporal data to improve predictive accuracy, as well as by optimizing computational efficiency through better algorithms or parallel processing. Including municipal planning constraints and demographic layers would also make the system more practical for real-world use. Future efforts will focus on validating the framework across different cities and climate zones, extending it to account for long-term urban growth and climate change, and developing it into a modular, interoperable platform that integrates seamlessly with existing geospatial and urban analytics tools—ultimately supporting scalable, sustainable, and data-informed UHI mitigation strategies.

REFERENCES

[1] T. R. Oke, "The energetic basis of the urban heat island," *Quarterly Journal of the Royal Meteorological Society*, vol. 108, no. 455, pp. 1–24, 1982, doi:10.1002/qj.49710845502.

[2] J. A. Voogt and T. R. Oke, "Thermal remote sensing of urban climates," *Remote Sensing of Environment*, vol. 86, no. 3, pp. 370–384, 2003, doi:10.1016/S0034-4257(03)00079-8.

[3] D. Zhou, Z. Yang, Q. Huang, S. Li, J. Wang, and X. Wu, "Remote sensing of urban heat islands: Progress, challenges, and perspectives," *Remote Sensing*, vol. 11, no. 1, pp. 1–26, 2019, doi:10.3390/rs11010048.

[4] L. Zhao, X. Lee, R. B. Smith, and J. Oleson, "Interacting effects of climate change and urbanization on heat stress in 2050," *Environmental Research Letters*, vol. 13, no. 3, 034003, 2018, doi:10.1088/1748-9326/aaa00e.

[5] I. Tobias, M. Wolf, A. Gouveia, J. D. Spadaro, P. C. Kinney, and R. H. Kjellstrom, "Attributable human mortality during extreme heat events: a European multi-city analysis," *Environmental Health Perspectives*, vol. 130, no. 5, 057001, 2022, doi:10.1289/EHP9987.

[6] S. Hajat, L. Ebi, and B. Kovats, "Heat-health warning systems: a comparison of the predictive capacity of different approaches to heat–health warnings," *Environmental Health Perspectives*, vol. 118, no. 1, pp. 134–142, 2010, doi:10.1289/ehp.0900908.

[7] Climate Central, "Urban Heat Hot Spots in 65 Cities," Jul. 2024. [Online]. Available: https://www.climatecentral.org/climate-matters/urban-heat-islands-2024. [Accessed: Oct. 25, 2025].

[8] Resources for the Future, "Urban Heat Islands 101," 2022. [Online]. Available: https://www.rff.org/publications/explainers/urban-heat-islands-101/. [Accessed: Oct. 25, 2025].

[9] United Nations, "68% of the world population projected to live in urban areas by 2050," UN DESA, 2018. [Online]. Available: https://www.un.org/development/desa/en/news/population/2018-revision-of-world-urbanization-prospects.html. [Accessed: Oct. 25, 2025].

[10] M. Santamouris, "Cooling the cities – A review of reflective and green roof mitigation technologies to fight heat island and improve comfort in urban environments," *Solar Energy*, vol. 103, pp. 682–703, 2014, doi:10.1016/j.solener.2012.07.003.

[11] M. Susca, S. R. Gaffin, and G. R. Dell'Osso, "Positive effects of vegetation: Urban heat island and green roofs," *Environmental Pollution*, vol. 159, nos. 8–9, pp. 2119–2126, 2011, doi:10.1016/j.envpol.2011.03.007.

[12] A. V. Georgescu, C. E. Lobell, P. K. Howarth, and T. O. Ahlquist, "Urban vegetation can save lives under future climate conditions," *Environmental Research Letters*, vol. 13, no. 5, 054014, 2018, doi:10.1088/1748-9326/aabc77.

[13] G. Manoli, T. X. Zuidema, D. Zhou, J. P. Pal, L. L. T. H. Nguyen, A. K. Salmond, R. J. Hall, and P. Röder, "Magnitude of urban heat islands largely

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explained by climate and population," *Nature*, vol. 573, pp. 55–60, 2019, doi:10.1038/s41586-019-1512-9.

[14] K. P. Nadgir, "Geo-Intelligent Solutions for Urban Heat Island (UHI) Detection and Mitigation Framework," *IEEE Access*, vol. 13, pp. 146676-146689, 2025, doi: 10.1109/ACCESS.2025.3599723

[15] G. Sundar, P. Patchaiammal, B. V. Venkateshwarulu, T. P. Kumar, K. Rajamannar, and R. K. Tripathi, "Integrative Remote Sensing Approaches Using Generative Adversarial Networks for Urban Heat Island Analysis and Mitigation," *Remote Sensing in Earth Systems Sciences*, vol. 7, pp. 681–698, 2024, doi: 10.1007/s41976-024-00156-6.

[16] F. Marando, M. P. Heris, G. Zulian, A. Udías, L. Mentaschi, N. Chrysoulakis, D. Parastatidis, and J. Maes, "Urban heat island mitigation by green infrastructure in European Functional Urban Areas," *Sustainable Cities and Society*, vol. 77, p. 103564, 2022, doi: 10.1016/j.scs.2021.103564.

[17] X. Li, W. Lu, Z. Peng, Y. Zhang, and J. Huang, "Generative design of walkable urban cool spots using a novel heuristic GAN×GAN approach," *Building and Environment*, vol. 266, p. 112027, 2024, doi: 10.1016/j.buildenv.2024.112027.

[18] Y. Wang, U. Berardi, and H. Akbari, "Comparing the effects of urban heat island mitigation strategies for Toronto, Canada," *Energy and Buildings*, vol. 114, pp. 2–19, 2016, doi:10.1016/j.enbuild.2015.06.046.

[19] E. Elmarakby and H. Elkadi, "Prioritising urban heat island mitigation interventions: Mapping a heat risk index," *The Science of The Total Environment*, vol. 948, p. 174927, 2024, doi: 10.1016/j.scitotenv.2024.174927.

[20] O. Y. A. Mohamed and I. Zahidi, "Artificial intelligence for predicting urban heat island effect and optimising land use/land cover for mitigation: Prospects and recent advancements," *Urban Climate*, vol. 55, p. 101976, 2024, doi: 10.1016/j.uclim.2024.101976.

[21] S. Dolhopolov, T. Honcharenko, I. Sachenko, and D. Gergi, "Enhancing Urban Planning with LoRa and GANs: A Project Management Perspective," *5th International Workshop on IT Project Management*, Bratislava, Slovakia, May 22, 2024, vol. 3709, paper 19. [Online]. Available: https://ceur-ws.org/Vol-3709/paper19.pdf. [Accessed: Oct. 25, 2025].

[22] W. Cheng, Y. Chu, C. Xia, B. Zhang, J. Chen, M. Jia, and W. Wang, "UrbanGenoGAN: pioneering urban spatial planning using the synergistic integration of GAN, GA, and GIS," *Frontiers in Environmental Science*, vol. 11, Dec. 2023, doi: 10.3389/fenvs.2023.1287858.

[23] Y. Zhang, J. Ge, S. Wang, and C. Dong, "Optimizing urban green space configurations for enhanced heat island mitigation: A geographically weighted machine learning approach," *Sustainable Cities and Society*, vol. 119, 106087, Jan. 2025, doi:10.1016/j.scs.2024.106087.

[24] D. Li, E. Bou-Zeid, and M. Oppenheimer, "The Effectiveness of Cool and Green Roofs as Urban Heat Island Mitigation Strategies," *Environmental Research Letters*, vol. 9, no. 5, 055002, 2014, doi: 10.1088/1748-9326/9/5/055002.

[25] J. Chen, R. Shi, G. Sun, Y. Guo, M. Deng, and X. Zhang, "Simulation-Based Optimization of the Urban Thermal Environment through Local Climate Zones Reorganization in Changsha City, China with the FLUS Model," *Sustainability*, vol. 15, no. 16, 12312, 2023, doi:10.3390/su151612312.

[26] W. Gan, Z. Zhao, Y. Wang, Y. Zou, S. Zhou, and Z. Wu, "UDGAN: A New Urban Design Inspiration Approach Driven by Using Generative Adversarial Networks," *Journal of Computational Design and Engineering*, vol. 11, no. 1, pp. 305–324, 2024, doi:10.1093/jcde/qwae014.

[27] R. Tian, "Suggestive Site Planning with Conditional GAN and Urban GIS Data," in *Proceedings of the 2020 DigitalFUTURES*, pp. 103–113, Jan. 2021, https://doi.org/10.1007/978-981-33-4400-6_10.

[28] P. F. Yuan, J. Yao, C. Yan, X. Wang, and N. Leach, Eds., *Proceedings of the 2020 DigitalFUTURES: The 2nd International Conference on Computational Design and Robotic Fabrication (CDRF 2020)*, Springer, Singapore, 2021, doi:10.1007/978-981-33-4400-6.

[29] A. Yadav and J. Singh, "A Study on Urban Heat Island (UHI): Challenges and Opportunities for Mitigation," *Current World Environment*, vol. 19, no. 1, pp. 436–453, 2024, doi:10.12944/CWE.19.1.37.

[30] A.-A. Kafy, N. N. Dey, A. A. Rakib, Z. A. Rahaman, N. M. R. Nasher, and A. Bhatt, "Modeling the relationship between land use/land cover and land surface temperature in Dhaka, Bangladesh using CA-ANN algorithm,", *Environmental Challenges*, vol. 4, 100190, 2021, https://doi.org/10.1016/j.envc.2021.100190.

[31] A. K. Tiwari and R. Kanchan, "Analytical study on the relationship among land surface temperature, land use/land cover and spectral indices using geospatial techniques,", *Discover Environment*, vol. 2, Article 1, 2024, https://doi.org/10.1007/s44274-023-00021-1.

[32] Sanchez-Lengeling, et al., "A Gentle Introduction to Graph Neural Networks", Distill, 2021.

[33] M. Pesaresi and P. Panagiotis, "GHS-BUILT-C R2023A - GHS settlement characteristics, derived from Sentinel-2 composite (2018) and other GHS R2023A data," *European Commission, Joint Research Centre (JRC)*, 2023. [Dataset]. doi: 10.2905/3C60DDF6-0586-4190-854B-F6AA0EDC2A30.

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